TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Capstone Design Project I

Academic Year 2026–2027

PerDrobe Project Proposal

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Project name: PerDrobe

Project webpage: https://beratergun.github.io/perdrobe/

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Project Description

PerDrobe is a proposed personalized wardrobe, outfit and clothing recommendation platform for TED University's CMPE 491–492 capstone project. It addresses the difficulty of turning style inspiration into usable outfits and selecting suitable catalogue items without overlooking clothes a user already owns. The planned system will maintain a private digital wardrobe and combine garment attributes with explicit style and fit preferences. Likes, saves and follows within a lightweight social discovery layer will provide additional preference signals. Size guidance will use optional measurements, usual sizes and available product charts, and will be limited to cases with sufficient supporting information.

The project's technical focus is a recommendation pipeline for candidate generation, compatibility analysis and personalized ranking. Outfit Remix will adapt a discovered outfit through compatible substitutions from the user's wardrobe, while wardrobe-aware shopping will suggest additions that work with owned garments. We will compare a clothing-attribute baseline, a configuration that adds preferences and interaction signals, and a configuration that also uses wardrobe context. Planned evaluation covers ranking relevance, user-rated usefulness, diversity, coverage and response time, with size or fit agreement assessed only where reference labels exist. The application and experiments are planned work; no results are claimed. Duo Match and image-based extensions remain conditional on completion of the core system.